

Reconstructing Curves Using the Crust

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March 19, 2026

Polygonal Reconstruction

- Points are taken from a smooth curve, known as *sampling*
- Segments are drawn to connect all adjacent samples, and connect no other points
- Under the correct sampling conditions, algorithms can guarantee a successful reconstruction

Geometric Tools

Triangulations

- A triangulation creates a mesh that connects all points in a set using only triangles
- Any set of points can be triangulated, usually in many different ways
- Some triangulations are more useful than others

Delaunay Triangulations

- Delaunay triangulations obey the **empty circle property**
- Creates a uniform triangulation, reducing the number of thin triangles
- Always exists and is unique for a given set of points

Voronoi Diagrams

- Divide the plane into *regions of proximity*
- Consists of Voronoi edges and Voronoi vertices
- Useful to quickly find the nearest point in the set anywhere in the plane

The Delaunay-Voronoi Relationship

- Voronoi vertices are equidistant to at least 3 points in the set
- Circles centered at Voronoi vertices obey the empty circle property
- Connecting these points yields a triangle in the Delaunay triangulation
- The Voronoi diagram and the Delaunay Triangulation are *dual graphs* of one another